Part E · Justification for the clinical review figure

Source: vitals_10k.csv, 10,000 synthetic patient records generated for CDB601220 Week 3.

1 · The four panels

Panel	Question it answers	Why this chart	Week 2 channel rule
1 Histogram	Is systolic pressure one population or several?	One continuous variable, shape is the question; a histogram shows modes that a mean or box would hide.	Position on a common aligned scale carries the count; hue only labels the two modes and never carries a value.
2 Box plot	Which ward keeps patients longest?	One skewed measure across four categories: median, spread and outliers per ward, side by side.	Aligned position on one log axis; categories ordered by median so rank reads left to right.
3 Scatter	How strongly does cholesterol follow age?	Two continuous variables: a scatter shows form, strength and spread. Panel 4 pointed to this pair.	Two aligned position scales; sex is carried by shape *and* colour (redundant coding); alpha tackles overplotting.
4 Heatmap	Which variable pairs move together?	A screening tool for 36 pairs at once; it points, it does not prove.	Colour is a weaker channel, so cells are annotated; hue = sign, lightness = size, diverging, midpoint 0.

2 · Every parameter I chose

Panel 1: bin width **3 mmHg**, aligned to whole mmHg (Freedman–Diaconis gives about 3.8; 3 keeps both peaks and a clean valley). Split at the density valley, **141 mmHg**, found from a KDE rather than by eye. **Panel 2: $\log_{10}$ y-axis** (length of stay is strongly right-skewed); **showfliers = True**; wards sorted by median; the 2 zero-day stays cannot be drawn on a log axis. **Panel 3:** all 10,000 points, **alpha 0.18**, marker size 9, OLS line, no subsampling. **Panel 4: cmap RdBu_r, vmin = −1, vmax = +1, center = 0**, upper triangle masked, annotation .2f, pairwise-complete correlations, map_mmhg excluded. **Figure:** 12.6 × 8.4 in, **300 dpi**, bbox tight, house palette (#0072B2, #D55E00, #009E73), spines off.

3 · Every change to the data, with rule, count and effect

Rule	Count	Effect
ward: 12 spellings conformed to 4 (strip, lower-case, map)	12 → 4; 0 unmapped	Groupby no longer splits wards
admit_date: 3 formats parsed explicitly; slash dates are day-first	3,382 / 3,365 / 3,253; 0 unparsed	mm/dd would be impossible for 1,933 rows; range 1 Jan–30 Apr 2026
bmi 999 (not recorded) and 0 (impossible) set to missing	402 + 84 = 486	mean BMI 66.2 → 27.3 kg/m²
heart_rate > 250 bpm removed as device faults; < 30 bpm (all 0, flat-line) removed	29 + 8 = 37	mean 78.8 → 78.0 bpm
los_days > 30 d set to missing (nothing lies between 24.0 and 139.8 d)	18	mean 2.31 → 1.95 d; median 1.38 → 1.38
map_mmhg excluded from heatmap: derived from systolic and diastolic	1 column	r with systolic 0.97 is arithmetic

The ethics clause

Readings above 250 bpm were removed as device faults (n = 29). Mean heart rate moves from 78.8 to 78.0 bpm. The figure was also produced without the exclusion; it is in the appendix (partD_appendix_heart_rate.png). The same applies to the 8 flat-line (0 bpm) readings. Rows were never silently dropped: faulty *values* were set to missing, so each patient still counts everywhere else, and no box plot hides its outliers (showfliers = True).

Symmetry test. Would I have made the same removals if they had weakened my conclusion? Yes. Each rule was fixed by physiology or by the file's own coding (no human heart beats at 400 bpm; 999 is a missing-value code; no inpatient stays 270 days when nothing lies between 24 and 140) *before* I looked at any relationship. Checking afterwards: the headline correlation, age and cholesterol, is r = 0.34 before and 0.34 after cleaning, because neither column was touched; the ward medians for length of stay are unchanged (overall 1.38 → 1.38 d). The removals change means, not conclusions, so they are cleaning rules, not edits.

Sources: Tufte (1983); Wilke, *Fundamentals of Data Visualization*; Week 2 channel-ranking notes; seaborn and SciPy documentation.
All data synthetic.